Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP(01CT0513)	**Aim: Design FIR filter with windowing method.**	
Experiment 6	**Date: 27/09/2026**	**Enrollment No: 92510133048**

AIM: Design FIR filter with windowing method.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

# Experiment 6: FIR Filter Design Using Windowing Method

N = 31
Fs = 10000
window_type = 'hann'

# 1. FIR Low-Pass Filter
b_lpf = firwin(
    numtaps=N,
    cutoff=3000,
    window=window_type,
    pass_zero='lowpass',
    fs=Fs
)

# 2. FIR High-Pass Filter
b_hpf = firwin(
    numtaps=N,
    cutoff=3000,
    window=window_type,
```

```
    pass_zero='highpass',
    fs=Fs
)

# Sampling frequency for Band-Pass and Band-Stop filters
Fs_band = 20000

# 3. FIR Band-Pass Filter
b_bpf = firwin(
    numtaps=N,
    cutoff=[3000, 6000],
    window=window_type,
    pass_zero='bandpass',
    fs=Fs_band
)

# 4. FIR Band-Stop Filter
b_bsf = firwin(
    numtaps=N,
    cutoff=[3000, 6000],
    window=window_type,
    pass_zero='bandstop',
    fs=Fs_band
)

# Filter list
```

```
filters = [
    ("FIR Low-Pass Filter (Fc = 3 kHz)", b_lpf, Fs),
    ("FIR High-Pass Filter (Fc = 3 kHz)", b_hpf, Fs),
    ("FIR Band-Pass Filter (3-6 kHz)", b_bpf, Fs_band),
    ("FIR Band-Stop Filter (3-6 kHz)", b_bsf, Fs_band)
]

# Plot frequency responses
plt.figure(figsize=(12, 9))

for i, (title, b, fs) in enumerate(filters, start=1):
    f, H = freqz(b, worN=2048, fs=fs)

    magnitude_db = 20 * np.log10(
        np.maximum(np.abs(H), 1e-10)
    )

    plt.subplot(2, 2, i)
    plt.plot(f / 1000, magnitude_db)
    plt.title(title)
    plt.xlabel("Frequency (kHz)")
    plt.ylabel("Magnitude (dB)")
    plt.grid(True)

plt.tight_layout()
plt.show()
```

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```
# Display filter coefficients
print("LOW-PASS FILTER COEFFICIENTS:")
print(b_lpf)

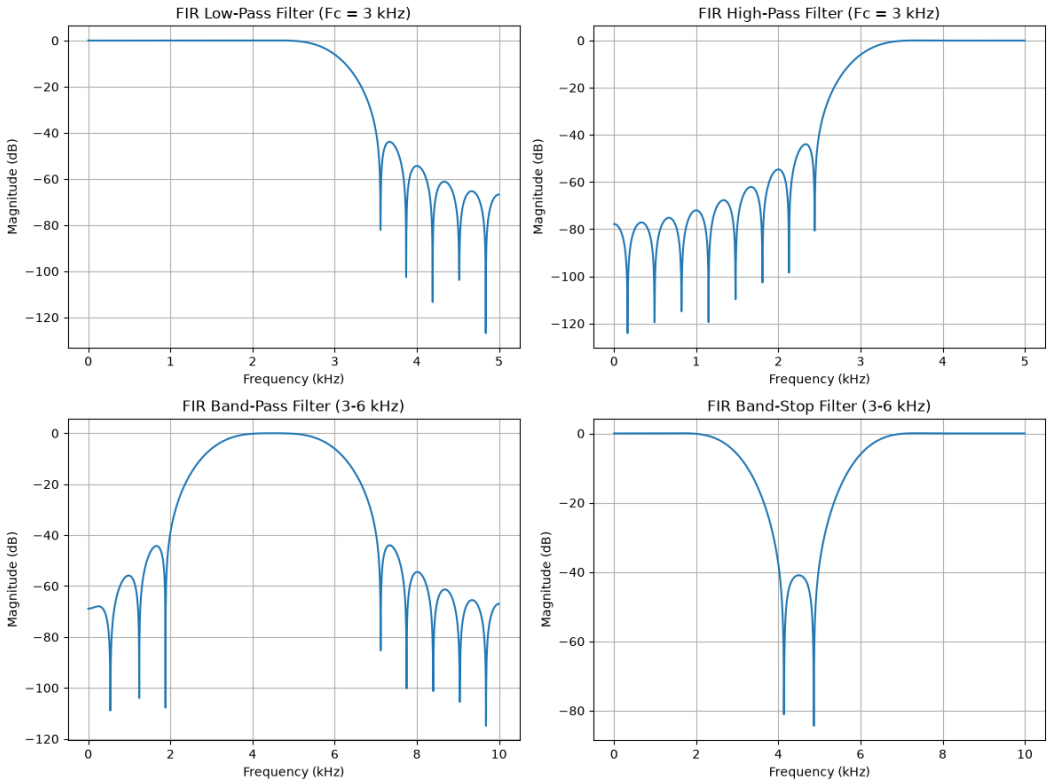
print("\nHIGH-PASS FILTER COEFFICIENTS:")
print(b_hpf)

print("\nBAND-PASS FILTER COEFFICIENTS:")
print(b_bpf)

print("\nBAND-STOP FILTER COEFFICIENTS:")
print(b_bsf)
```

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Output:




```
LOW-PASS FILTER COEFFICIENTS:
[ 0.00000000e+00  2.36294439e-04 -6.22213220e-04 -1.48904646e-03
  4.55351027e-03 -5.84801114e-18 -1.16227064e-02  1.04726495e-02
  1.47629668e-02 -3.30275566e-02  1.75440334e-17  6.31702710e-02
 -5.64178020e-02 -8.95165958e-02  2.99461575e-01  6.00077307e-01
  2.99461575e-01 -8.95165958e-02 -5.64178020e-02  6.31702710e-02
  1.75440334e-17 -3.30275566e-02  1.47629668e-02  1.04726495e-02
 -1.16227064e-02 -5.84801114e-18  4.55351027e-03 -1.48904646e-03
 -6.22213220e-04  2.36294439e-04  0.00000000e+00]

HIGH-PASS FILTER COEFFICIENTS:
[ 0.00000000e+00 -2.36155069e-04  6.21846227e-04  1.48816819e-03
 -4.55082453e-03 -3.89637458e-18  1.16158511e-02 -1.04664726e-02
 -1.47542593e-02  3.30080763e-02  1.16891238e-17 -6.31330120e-02
  5.63845257e-02  8.94637973e-02 -2.99284947e-01  3.99815580e-01
 -2.99284947e-01  8.94637973e-02  5.63845257e-02 -6.31330120e-02
  1.16891238e-17  3.30080763e-02 -1.47542593e-02 -1.04664726e-02
  1.16158511e-02 -3.89637458e-18 -4.55082453e-03  1.48816819e-03
  6.21846227e-04 -2.36155069e-04  0.00000000e+00]

BAND-PASS FILTER COEFFICIENTS:
[-0.00000000e+00  8.94419399e-05 -2.92433322e-04  9.11976393e-04
  8.35090976e-03 -8.69285507e-18 -2.13154614e-02 -6.41404375e-03
  6.93843088e-03 -1.25015584e-02  4.73216989e-02  1.01289253e-01
 -8.53015953e-02 -2.32242569e-01  4.43207365e-02  2.97331003e-01
...
 -4.47023710e-02  2.34242350e-01  8.60361054e-02 -1.02161429e-01
 -4.77291739e-02  1.26092061e-02 -6.99817593e-03  6.46927343e-03
  2.14990033e-02 -9.74189668e-19 -8.42281730e-03 -9.19829187e-04
  2.94951391e-04 -9.02121014e-05  0.00000000e+00]
```